Clinical Decision Support System

MRI Brain Tumor Analysis Report

Patient ID	MRI-3_2207D1
Name	Anonymous
Age	56 years
Sex	Not specified
Comorbidities	None reported
Symptoms	None reported
Date/Time	2026-05-21T08:08:13.733544
Run ID	20260521_080813_2207d1
Tumor Type	MENINGIOMA
Confidence	99.9%
Radiologist	Approved

Clinical Summary

Meningioma detected. Typically an extra-axial, slow-growing benign tumor. Assessment of surgical candidacy, symptom burden, and proximity to critical structures is required before deciding between observation and intervention.

Recommended Next Steps

1. Neurosurgery consultation
2. CT to evaluate calcification and bone involvement
3. MR or CT angiography if adjacent to major venous sinuses
4. Ophthalmology referral if near optic chiasm
5. Follow-up MRI at 3-6 months if watchful waiting is chosen

Clinical Guidelines

MENINGIOMA
Definition: Meningioma is a tumor arising from the arachnoid cap cells of the meninges — the protective membranes surrounding the brain and spinal cord. It is extra-axial (outside brain parenchyma).
Characteristics:
- Usually slow-growing; most are WHO Grade I (benign)
- Account for approximately 37% of primary brain tumors
- More common in women (2:1 female-to-male ratio)
- Associated with NF2 mutation, prior radiation exposure
- Can compress adjacent brain tissue despite being benign
- Atypical (Grade II) and malignant (Grade III) variants exist (~15-20%)

Typical Imaging Findings (MRI):
- Well-circumscribed, round or lobulated extra-axial mass
- Homogeneous isointense or hypointense on T1
- Homogeneous intense enhancement with gadolinium
- Dural tail sign (thickening/enhancement of adjacent dura) — pathognomonic
- Calcification visible on CT (20-30% of cases)
- Minimal surrounding edema (unless large or aggressive)
- Hyperostosis (bone thickening) adjacent to skull-base meningiomas

Recommended Next Steps:
1. CT scan to evaluate calcification and bone involvement
2. MRI with and without contrast to characterize extent and dural involvement
3. CT or MR angiography if near major venous sinuses
4. Neurosurgery consultation to assess resectability and symptom burden
5. Observation with 6-month follow-up MRI for small asymptomatic tumors
6. Ophthalmology referral if near optic apparatus
7. Radiation oncology referral for surgery-ineligible patients or recurrence

Prognosis:
- Grade I (benign): excellent prognosis; 10-year recurrence rate ~10% after gross total resection
- Grade II (atypical): 10-year recurrence ~50%
- Grade III (malignant): aggressive course, poor prognosis similar to high-grade glioma
- Simpson grade of resection strongly predicts recurrence

Treatment:
- Watchful waiting for small, asymptomatic, incidental meningiomas
- Surgical resection (Simpson Grade I-II) for symptomatic tumors
- Stereotactic radiosurgery (SRS/Gamma Knife) for small...